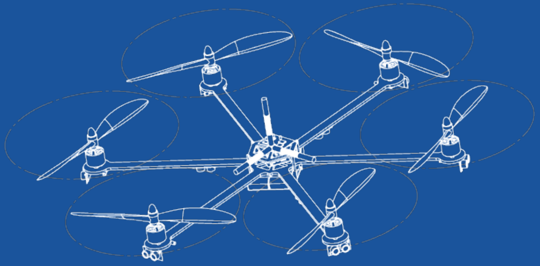
Universität Stuttgart
Institut für Flugmechanik und Flugregelung



Region of attraction with integral quadratic constraints

System Theoretical Methods for Flight Control

Saturday 16th May, 2026

First Last

Overview

Goal of the Paper

Goal of the Paper: designing safe "instructions" (control laws) for complex machines or robots using only recorded experimental data

General Approach

General Approach: Typischerweise wird Sicherheit mit Barrierefunktionen zertifiziert.

New Approach

New Approach:

- Anstatt Barriere-Funktionen, werden Dichtefunktionen benutzt
- Führt zu einem konvexen Problem.
- -> SDP

Overlays

1. Some people like overlays in their presentation.
2. i.e., slides that build up step-wise.
3. Note that these overlays do not increase the number of slides on right bottom corner.
4. Some commands include ‘*pause*’, ‘*uncover*’, ‘*only*’, etc.
5. There is no need to copy the information from one slide to the other to create overlays!

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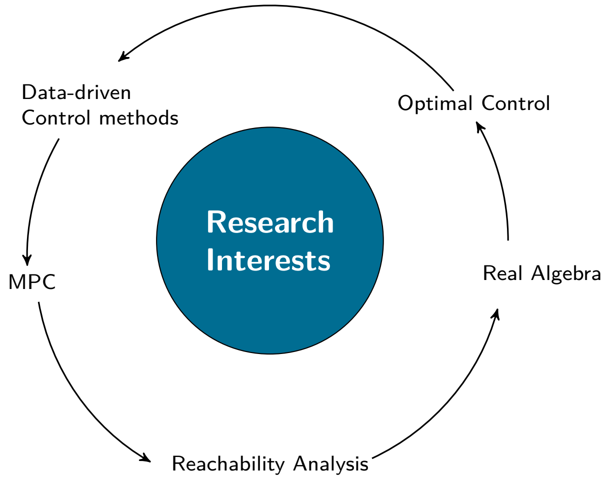
In the next slides, we will see some examples.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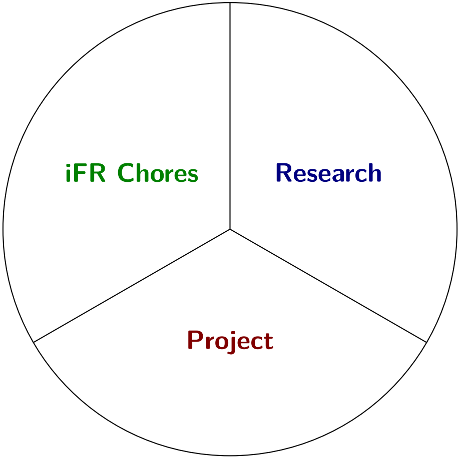
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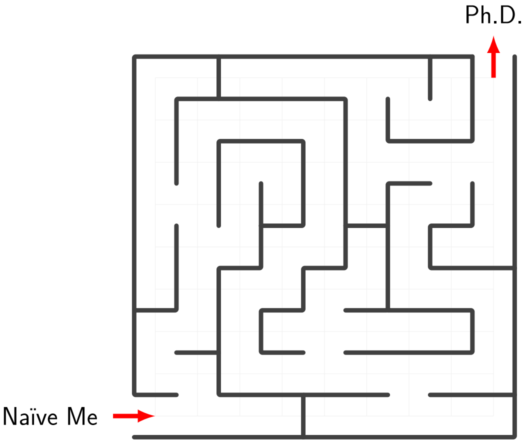
Research
Project

TikZ Example - 3



TikZ Example - 4

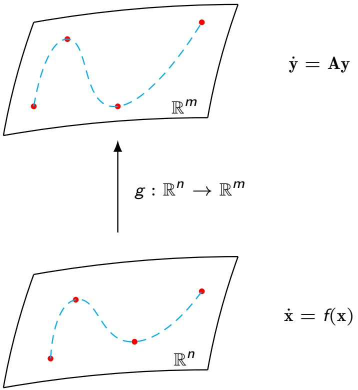


Figure 1: Some description

TikZ Keviats

Well, tikz is AWESOME, however as you probably noticed by now, it can take a lot of time compiling the whole document if it contains multiple tikzpictures.
If possible, generate the tikzpictures in a standalone file, i.e., generate .pdf files with the figures and only import those to the main file. Believe me, it will reduce by a lot the time you spend waiting for the compiled file.

Equations

Since this is latex, we can use equations.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{1}$$

All equations are numbered by the order they appear in the presentation. But if you don't want a numbered equation, you can still do it

$$E = mc^2$$

Blocks

One useful and nice feature is the block environment (*newtheorem*). It can be used to highlight important information. (see preamble.tex)

Conferences

1. *American Control Conference, Toronto, July 2024*
   Initial deadline: ***September 22, 2023***
2. *European Control Conference, Stockholm, June 2024*
   Initial deadline: ***October 25, 2023***

Theorem (Theorem name)

This is a theorem.

Remark

This is a remark.

Include figures

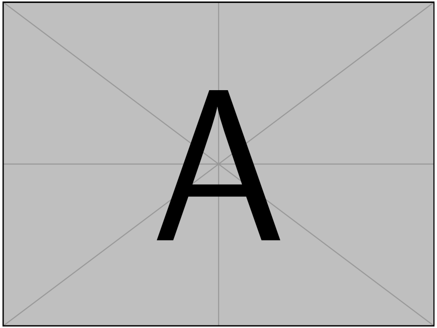


Figure 2: Caption 1

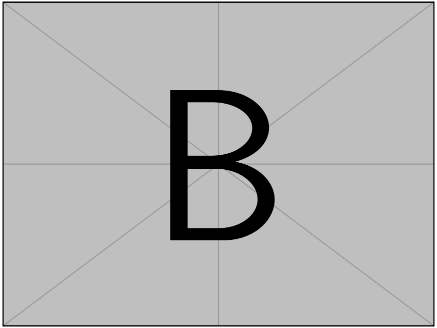
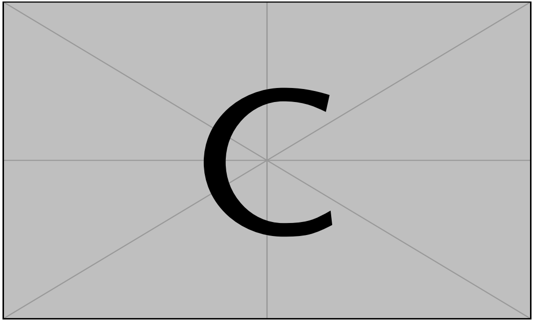


Figure 3: Caption 2

Insert videos



Note that only some pdf viewer can render the video. For presentions I recommend using '*pdfpc*', an open source presentation viewer that mimics the presentation layout from PowerPoint (some advantages include a timer, a way to see the next slide before it is presented and the use of a pointer or a pen)

Citations

Just as a normal .tex document we can use .bib files to cite references.

- Some random text... [1]
- Another random text... [2, 3]

The references are shown at the end of the presentation automatically.

Bibliography

In the next slide, you have the default final slide. You can edit it as you see fit.
In the current state you just need to change the photo and a few information details.

Universität Stuttgart
Institute of Flight Mechanics and Controls

Thank you!

First Last

E-Mail: ...@ifr.uni-stuttgart.de

References I

[1] Russell A. Smith. Some Applications of Hausdorff dimension inequalities for ordinary differential equations. *Proc. Royal Society of Edinburgh: Section A Mathematics*, 104(3-4):235–259, 1986.

[2] Pablo A. Parrilo. Semidefinite programming relaxations for semialgebraic problems. *Mathematical Programming, Series B*, 96(2):293–320, 2003.

[3] James S. Muldowney. Compound matrices and ordinary differential equations. *Rocky Mountain Journal of Mathematics*, 20(4):857–872, 1990.